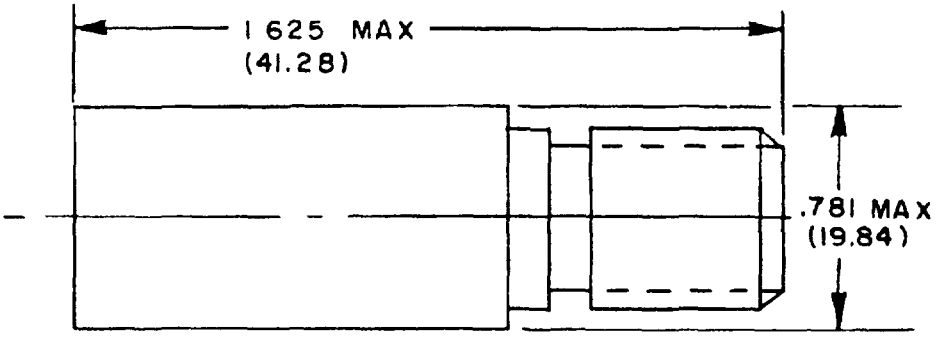
MILITARY SPECIFICATION SHEET

# CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY (SERIES N (CABLED) - SOCKET CONTACT, CLASS 2) FOR SEMIRIGID CABLE

This specification is approved for use by all Departments and Agencies of the Department of the Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012



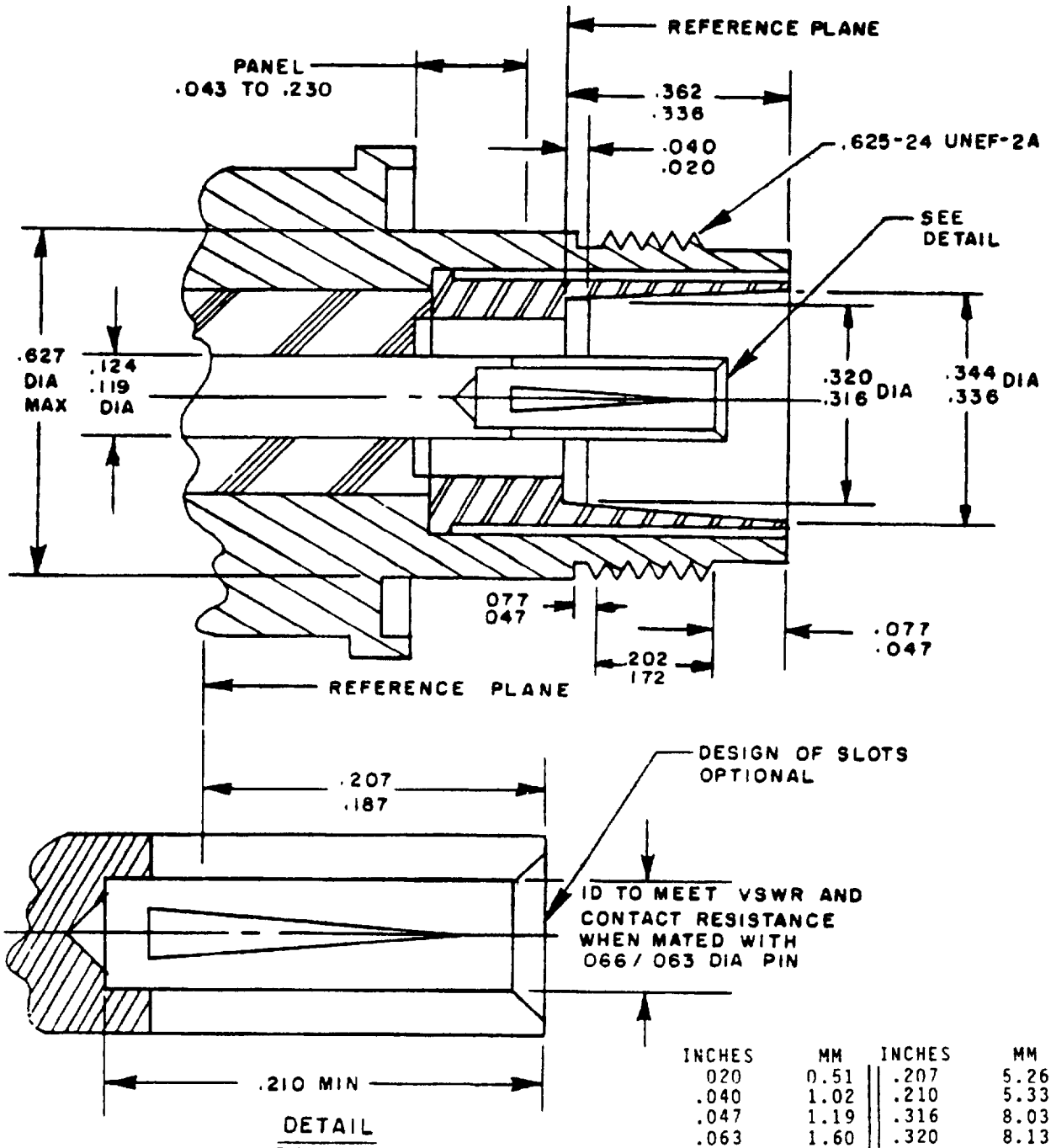

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only
3. Metric equivalents are in parentheses.
4. Dimension .781 (19.84 mm) is the largest overall diameter of the connector.
5. Wrench flats, when applicable, are to accommodate standard wrench opening per FED-STD-H28.
6. All undimensioned pictorial representations are for reference purposes only.
7. Dimension 1.625 (41.28 mm) defines the maximum length of the connector when assembled to the appropriate cable.

FIGURE 1 General configuration



FSC 5935

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| INCHES | MM | INCHES | MM |
|---|---|---|---|
| 020 | 0.51 | .207 | 5.26 |
| .040 | 1.02 | .210 | 5.33 |
| .047 | 1.19 | .316 | 8.03 |
| .063 | 1.60 | .320 | 8.13 |
| .066 | 1.68 | .336 | 8.53 |
| .077 | 1.96 | .344 | 8.74 |
| .119 | 3.02 | .356 | 9.04 |
| 124 | 3.15 | .362 | 9.19 |
| .172 | 4 37 | .422 | 10.72 |
| .202 | 5.13 | .627 | 15.93 |

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Mating dimensions for socket terminations.



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TABLE I. Dash number and applicable cable.

| Dash No. | Applicable cable 2/ |
|---|---|
| CATEGORY A - FIELD SERVICEABLE 1/ (NO SPECIAL TOOLS REQUIRED) | |
| 0001<br>3001 | M17/130-RG402* 0.141 dia. (3.58 mm)<br>M17/130-00001, 00002, 00003 |
| 0002<br>3002 | M17/129-RG401* 0.250 dia. (6.35 mm) |
| CATEGORY E - FIELD REPLACEABLE 1/ (STANDARD ASSEMBLY TOOL KIT) | |
| 0003<br>3003 | M17/130-RG402* 0.141 dia. (3.58 mm)<br>M17/130-00001, 00002, 00003 |

| Dash No. | Applicable cable 2/ | Tool No. | Positioning dies | Locator pins | Crimp dies M22520/5 |
|---|---|---|---|---|---|
| CATEGORY F - FIELD REPLACEABLE MIL-C-22520 COMPRESSION TOOL) 1/ | | | | | |
| 3201 | M17/129-RG401* | M22520/ 36-01 | M22520/36-XX | M22520/36-XX | --- |
| 3202 | M17/130-RG402*<br>M17/130-00001, 00002, 00003 | | | | |
| 3203 | M17/129-RG401* | M22520/ 5-01 | --- | --- | -XX |
| 3204 | M17/130-RG402*<br>M17/130-00001, 00002, 00003 | | | | |

1/ These connectors have captivated center contacts.
2/ The latest version of each cable shall be applicable.
* Cable to be used when performing tests requiring cable.

NOTE. Direct solder type connectors shall not be used on cable larger than .141 nominal diameter.

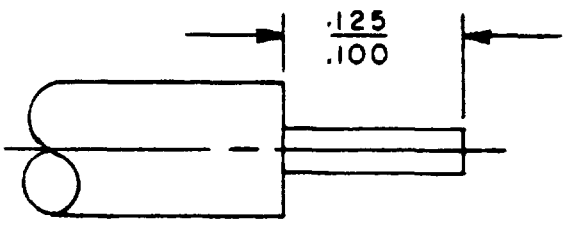



FIGURE 3. Cable stripping dimensions for category E and category F connectors.



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ENGINEERING DATA

Nominal impedance 50 ohms.

Frequency range 0 to 18 GHz

Voltage rating

500 volts rms maximum working voltage at sea level.

125 volts rms maximum at 70,000 feet.

Temperature rating:

-65°C to +105°C for M17/129-RG401.

-65°C to +105°C for M17/130-RG402, M17/130-00001, 00002, 00003.

REQUIREMENTS.

Dimensions and configuration See figure 1

Force to engage and disengage.

Longitudinal force: Not applicable.

Torque. 6 inch-pounds maximum.

Inspection conditions For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 6 to 10 inch-pounds.

Mating characteristics See figure 2 for dimensions.

Center contact (socket).

| | |
|---|---|
| Oversize test pin | 0.074 diameter minimum (non-closed entry contact only). |
| Insertion depth | .125 minimum |
| Number of insertions | 1. |

Insertion force test Steel test pin diameter .066 minimum.

| | |
|---|---|
| Test pin finish | 16 microinches. |
| Insertion force | 2 pounds maximum. |
| Insertion depth | .125 minimum. |

Withdrawal force test: Steel test pin diameter .063 maximum.

| | |
|---|---|
| Withdrawal force. | 2 ounces minimum. |
| Test pin finish | 16 microinches. |

Insulation resistance Method 3003 of MIL-STD-1344, 5,000 megohms minimum.

Center contact retention (applicable to captivated center contact connectors only) 6 pounds minimum axial force.

Corrosion (salt spray) Method 1001 of MIL-STD-1344, test condition B.

Voltage standing wave ratio (VSWR) From 2.0 to 18.0 GHz, 1.15 +.012 X freq. (GHz).



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Swept frequency VSWR test setup·

Item 6: VSWR shall be less than 1.008 +.002 F (F in GHz).

Item 16 VSWR shall be less than 1.008 +.002 F (F in GHz).

Second step of VSWR checkout procedure VSWR shall be less than 1.012 + 004 F (F in GHz).

Group B inspection· VSWR shall be less than 1.024 +.007 F (F in GHz).

Qualification and group C inspection· VSWR shall not exceed 1.08.

Connector durability: Method 2016 of MIL-STD-1344.

500 cycles at 12 cycles per minute maximum·

The connector shall meet the mating characteristics and force to engage and disengage requirements

Contact resistance (in milliohms maximum):

| | Initial | After environment |
|---|---|---|
| Center contact | 1.5 | 2.0 |
| Outer contact | .2 | Not applicable |
| Outer contact (stainless steel) | 2.0 | Not applicable |
| Cable outer conductor to body | .05 | Not apolicable |
| Cable outer conductor to stainless steel body | 2.0 | Not applicable |

Dielectric withstanding voltage. Method 3001 of MIL-STD-1344, 1,500 volts rms minimum at sea level.

Vibration, high frequency· Method 2005 of MIL-STD-1344, test condition IV, except the method of mounting shall be approved by the qualifying activity, and 30 g peak will be applicable.

Shock Method 2004 of MIL-STD-1344, test condition G.

Thermal shock Method 1003 of MIL-STD-1344, test condition A, except test high temperature shall be +115°C.

Humidity Method 1002 of MIL-STD 1344. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level·

Altitude 70,000 feet.

| Cable (semirigid) | V rms |
|---|---|
| M17/129-RG401 | 500 |
| M17/130-RG402, M17/130-00001, 00002, 00003 | 375 |


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Custodians.
Army - CR
Navy - EC
Air Force - 85

Review activities.
Army - MI, EA
Navy - SH
Air Force - 11, 17, 80, 99
DLA - ES

User activities.
Army - AT, AV
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity
Army - CR

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